

Spinal muscular atrophy (SMA) is an autosomal recessive neuromuscular disease affecting 1 in 10,000 live births. The pathological hallmark of SMA is the loss of α -motor neurons in the spinal cord, which causes progressive paralysis and ultimately premature death in infants. SMA is caused by insufficient levels of survival motor neuron (SMN), a regulatory protein required for the maintenance of the spinal cord nerve cells that control voluntary muscle movement.

The SMN protein is generated from the *SMN1* and *SMN2* genes, which differ in sequence by only a single nucleotide in exon 7. In healthy individuals, the gene product of *SMN1* is the full-length (FL) SMN protein, and the gene products of *SMN2* are both the FL-SMN protein as well as a truncated and generally nonfunctional protein (SMN Δ 7) that is rapidly degraded. Approximately 90% of the *SMN2* gene product is SMN Δ 7, and only 10% is FL-SMN (Figure 1). In most patients with SMA, *SMN1* is mutated such that its protein product is nonfunctional or deleted. The disease phenotype arises because the *SMN2* gene cannot fully compensate for the deficiency in FL-SMN protein synthesis.

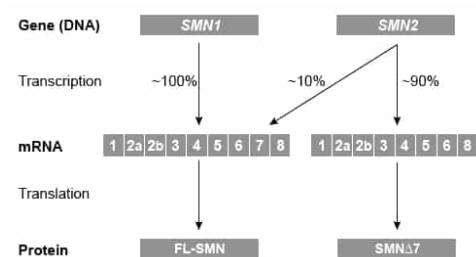


Figure 1 Structure of *SMN1* and *SMN2* transcripts

Antisense oligonucleotides (ASOs) that bind pre-mRNA were injected into SMA mice (*SMN1*^{-/-}, *SMN2*^{+/+}) exhibiting progressive tail necrosis, a characteristic measure of SMA severity in mice. ASO therapy effectiveness was assessed by measuring the tail length of wild-type (WT) mice and SMA mice exposed to increasing concentrations of ASOs or saline (Figure 2).

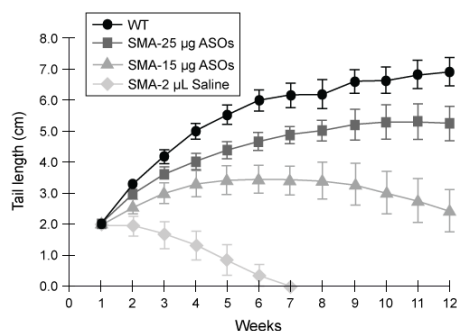


Figure 2 Tail length progression of WT mice compared to SMA mice treated with either saline or ASOs

The sequence below is a portion of exon 7 of the *SMN1* mRNA transcript:

5' – UCAAGUGAUUCUCCU – 3'

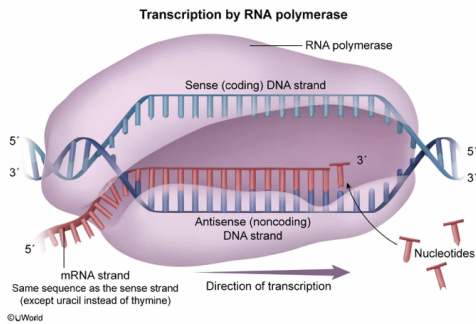
Which of the following represents the corresponding DNA coding strand sequence for this transcript?

- ☐ A. 5' – AGGAGAATCACTTGA – 3' [46%]
- ☐ B. 5' – TCCTCTTAGTGAAC – 3' [3%]
- ☐ C. 5' – AGTTCATAAGAGGA – 3' [18%]
- ☒ D. 5' – TCAAGTGATTCTCCT – 3' [32%]

Correct

32% answered correctly

Explanation:



DNA strands can be distinguished by their roles in transcription: One strand is known as the coding strand, and the other is the noncoding strand.

The **noncoding (antisense or template) DNA strand** provides the genetic information necessary for transcription of a new mRNA molecule by RNA polymerase II. Because the **newly generated mRNA transcript** is synthesized through base pairing with the noncoding strand, the nucleotide sequence of the mRNA strand is **complementary** to that of the noncoding strand, and its directionality is also **reversed**. In other words, the sequence of the noncoding strand read in the 5' to 3' direction is complementary to the sequence of the mRNA transcript read in the 3' to 5' direction. Note that **RNA molecules** contain uracil (U) nucleotides in place of thymine (T) nucleotides, which are found only in DNA.

The **coding (sense) strand** is the DNA strand that complements the noncoding DNA strand. Accordingly, the coding strand has the *same sequence and directionality* as the newly synthesized mRNA transcript with one exception (ie, the mRNA transcript has U rather than T nucleotides).

Therefore, the DNA coding strand sequence corresponding to the portion of the *SMN1* mRNA transcript shown in the question should have the *same* direction and sequence (with T in place of U) as the transcript. Only the strand represented in **Choice D** provides this correct coding sequence.

(Choice A) This DNA sequence, which is complementary and reversed relative to the mRNA sequence given, corresponds to the noncoding (template) strand.

(Choice B) The directionality of this sequence (ie, the designation of the 5' and 3' ends) is reversed, meaning this sequence is backward relative to the actual DNA coding strand sequence.

(Choice C) Although the directionality of this sequence is correct, the nucleotides are complementary to the nucleotides found in the actual DNA coding strand sequence.

Educational objective:

In transcription, RNA polymerase reads the noncoding DNA strand to produce a complementary mRNA transcript (with uracil replacing thymine). The sequence of the mRNA transcript is identical to the sequence of the coding DNA strand (again, with uracil instead of thymine).

Time spent: 0 sec
Subject:
Biology

QID: 400850
System:
Molecular Biology

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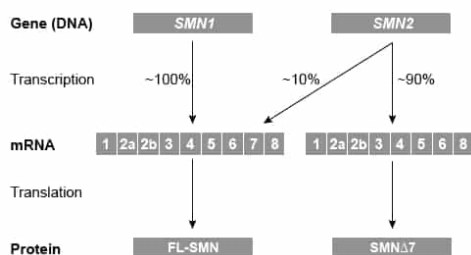


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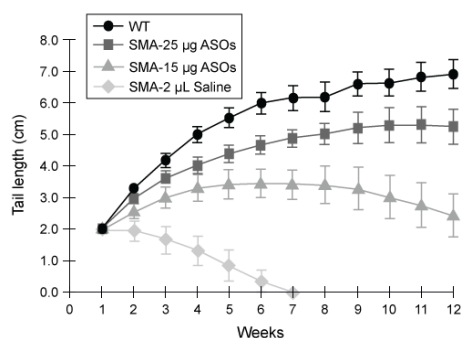


Figure 2 Tail length progression of WT mice compared to SMA mice treated with either saline or ASOs

Which step(s) are required to produce a mature FL-SMN transcript?

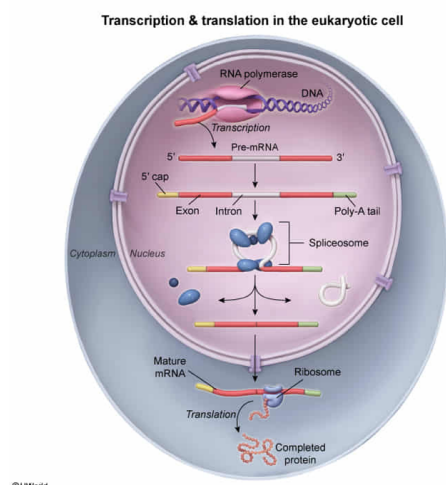
- I. Addition of multiple adenine nucleotides to the 3' end of *SMN1* pre-mRNA
- II. Removal of the 7-methylguanosine cap from mature mRNA
- III. DNA polymerase binding to the *SMN1* gene

- ✓ ☒ A. I only [41%]
☐ B. I and III only [34%]
☐ C. II and III only [10%]
☐ D. I, II, and III [13%]

Correct

41% answered correctly

Explanation:



Transcription is the process in which the nucleotide sequence of a DNA region is used as a template to synthesize a new RNA strand. Messenger RNA (mRNA) is the type of RNA molecule produced for the transcription of protein-expressing genes.

Transcription begins with the **binding** of **transcription factors** and **RNA polymerase II** to a specific AT-rich sequence (ie, TATA box) in the promoter region of double-stranded DNA. RNA polymerase II processes the noncoding DNA strand in a 3' to 5' direction until a termination sequence is recognized. As the enzyme travels down the DNA strand, it unravels the DNA double helix and relies on the complementary pairing of bases to catalyze the synthesis of a pre-mRNA strand, which grows in the 5' to 3' direction.

The pre-mRNA molecule generally undergoes the following **modifications** to be converted into mature mRNA:

1. A **7-methylguanosine (5' cap)** is **added** to the 5' end of pre-mRNA as the nascent transcript is being generated. This 5' cap is later recognized by the ribosome during translation and prevents the degradation of mRNA in the cytoplasm.
2. A chain of adenine nucleotides known as the **poly-A tail** is **added** to the **3' end** of the mRNA. This poly-A tail functions to prevent the mRNA from being degraded and facilitates the export of the mature mRNA from the nucleus to the cytoplasm (**Number I**).
3. In pre-mRNA, coding (exons) and noncoding (introns) regions are located between two **untranslated regions** (sequences that regulate **translation** but do not code for amino acids). The **noncoding** regions (introns) are **excised** from the pre-mRNA molecule, and the remaining untranslated and coding regions are joined via splicing.

(Number II) The 7-methylguanosine (5' cap) is added to (*not* removed from) pre-mRNA during processing to form mature mRNA.

(Number III) DNA polymerase catalyzes the synthesis of DNA molecules (ie, **replication**) and is *not* involved in transcription.

Educational objective:

Transcription is the process of synthesizing RNA from template DNA and begins with RNA polymerase II binding to the gene promoter region. RNA polymerase II reads the DNA strand in a 3' to 5' direction to generate a 5' to 3' pre-mRNA molecule. The pre-mRNA transcript undergoes 5' capping, the addition of a 3' poly-A tail, and excision of noncoding regions (introns) to be converted into mature mRNA.

Time spent:0 sec

Subject:

Biology

QID:400851

System:

Molecular Biology

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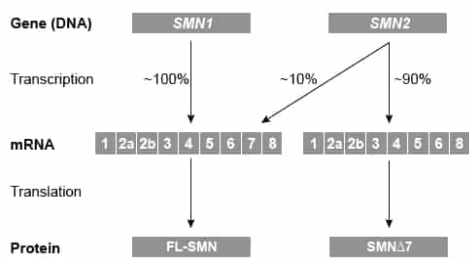


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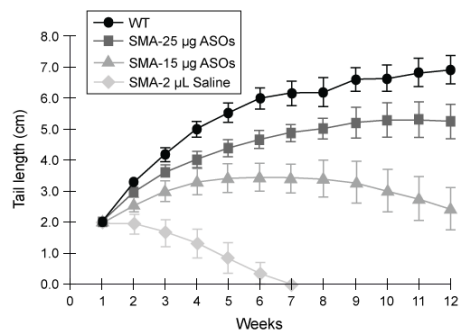
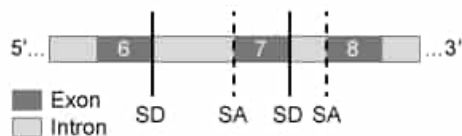


Figure 2 Tail length progression of WT mice compared to SMA mice treated with either saline or ASOs

Based on Figure 1, which of the following splicing mechanisms most likely contributes to SMA pathology? (Note: SD = splice donor site; SA = splice acceptor site.)

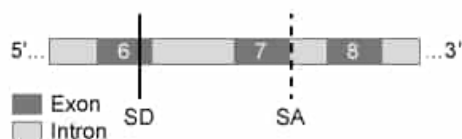
- ☐ A.
-
- [25%]
- ☒ B.
-
- [42%]

☐ C.



[21%]

☐ D.

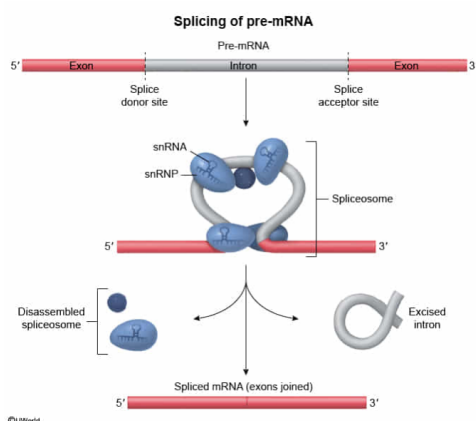


[10%]

Correct

42% answered correctly

Explanation:



Transcription in eukaryotes generates pre-mRNA molecules that contain **coding regions (exons)** and **noncoding regions (introns)**, the former of which contains the genetic information necessary for protein [translation](#). Nascent mRNA molecules undergo splicing, a process in which noncoding regions (introns) are excluded from the mature mRNA transcript. Splicing is driven by spliceosomes, complexes composed of specific proteins and small nuclear ribonucleoproteins (snRNPs) containing small nuclear RNA (snRNA).

In pre-mRNA, spliceosomes recognize specific sites in introns called splice donor sites and splice acceptor sites. Splice donor sites are located at the 5' end of the intron, next to the exon. In contrast, splice acceptor sites are found at the 3' end of the intron, adjacent to the exon. The **spliceosome** removes the intronic nucleotide sequence between donor and acceptor sites and **joins** the **exons** to produce a transcript with **no introns**.

As the passage shows, *SMN2* generates two different protein products through [alternative splicing](#), a process that allows various proteins to be generated from a single pre-mRNA transcript based on differential inclusion and exclusion of exons during splicing. Because 90% of the *SMN2* gene product is SMN Δ 7 (a truncated, nonfunctional version of FL-SMN), *SMN2* is unable to compensate for decreased FL-SMN expression when a loss-of-function mutation occurs in *SMN1*, as is the case in patients with SMA. The alternative splicing pathway that produces SMN Δ 7 rather than FL-SMN results from [exclusion of exon 7](#) from the mRNA transcript. Only Choice B shows splice donor and acceptor sites that would result in SMN Δ 7 production.

(Choices A and D) Figure 1 shows that the transcript for SMN Δ 7 *excludes* exon 7 in its entirety whereas the other exons *remain intact*. These options depict splice donor and acceptor sites that would remove only a portion of exon 7 or portions of other exons (ie, exon 6), and therefore do *not* represent splicing patterns resulting in SMN Δ 7 production.

(Choice C) This splicing pattern corresponds to synthesis of FL-SMN (ie, transcript with exon 7 *included*), which does *not* contribute to SMA. Specifically, FL-SMN is *depleted* in patients with SMA.

Educational objective:

The spliceosome removes introns from pre-mRNA by locating specific sequences within introns called 5' splice donor sites and 3' acceptor sites. Splice donor sites are located at the 5' end of the intron next to the exon, and splice acceptor sites are found at the 3' end of the intron adjacent to the exon.

Time spent:0 sec**Subject:**
Biology**QID:**400852**System:**
Molecular Biology

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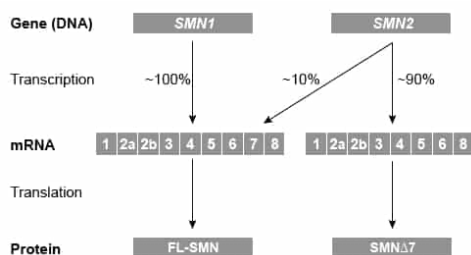


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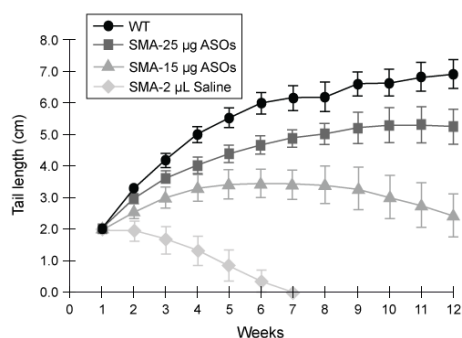


Figure 2 Tail length progression of WT mice compared to SMA mice treated with either saline or ASOs

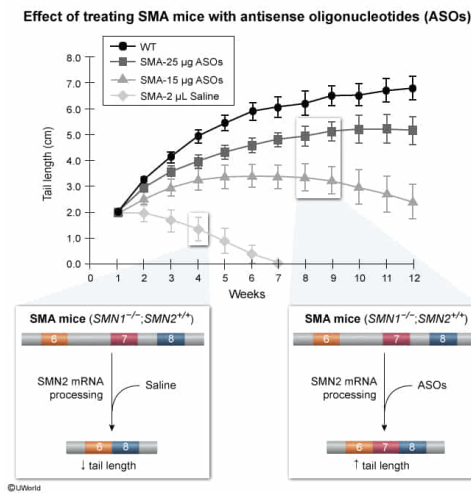
Based on the data shown in Figure 2, which statement best explains the function of antisense oligonucleotides (ASOs) in SMA mice? ASOs affect:

- ☐ A. transcription of *SMN2* pre-mRNA. [23%]
- ☒ B. processing of *SMN2* pre-mRNA. [49%]
- ☐ C. translation of *SMN2* mRNA. [21%]
- ☐ D. transport of *SMN2* mRNA to the cytoplasm. [4%]

Correct

50% answered correctly

Explanation:



According to the passage, **SMA mice** ($SMN1^{-/-}, SMN2^{+/+}$) are an animal model of SMA that produces insufficient FL-SMN. SMA mice exhibit progressive **tail necrosis**, which is seen as a loss of tail length and serves as a functional **measure of SMA severity**. In contrast, WT mice produce normal levels of FL-SMN transcripts, and therefore show no tail necrosis.

In this scenario, the effectiveness of antisense oligonucleotides (ASO) therapy was assessed by comparing the tail length of SMA mice following ASO or saline administration to that of untreated WT mice. Figure 2 shows that **ASO-treated SMA mice** have **increased tail length** compared to SMA mice treated with saline.

The passage states the ASOs bind pre-mRNA, meaning that a likely function of the ASOs used in this study is to affect the *processing* of pre-mRNA into mature mRNA. In particular, the improvement in tail length can likely be attributed to **ASO-induced inclusion** of exon 7 during **processing of SMN2 pre-mRNA**. Because *SMN2* produces FL-SMN (contains exon 7) and *SMNΔ7* (lacks exon 7), this exon 7 inclusion likely caused higher expression of FL-SMN compared to *SMNΔ7*. Consequently, ASO-treated mice had *increased* FL-SMN levels, and therefore *decreased* tail necrosis.

(Choice A) Figure 2 clearly shows that the ASOs used in the study alleviated one symptom associated with SMA in mice. As the passage indicates, SMA is associated with *increased* *SMNΔ7* production compared to FL-SMN production (ie, from having a nonfunctional *SMN1* gene and functional *SMN2* gene). Therefore, to improve SMA symptoms, ASO most likely alters the way *SMN2* pre-mRNA molecule is spliced, *increasing* FL-SMN expression. Splicing occurs *during pre-mRNA processing, not during transcription*. As the passage states, binding of ASOs to *pre-mRNA* indicates that ASOs act *after* transcription has already occurred.

(Choices C and D) The mature *SMN2* mRNA transcript (ie, 5' cap added, polyadenylated, and spliced) is transported from the nucleus to the cytoplasm, where translation (protein synthesis) occurs. As stated in the passage, ASOs only bind pre-mRNA, which *remains in the nucleus* until being processed into a mature transcript. As a result, ASOs would *not* affect transport of *mature SMN2* mRNA to the cytoplasm or its translation.

Educational objective:

Processing (which includes splicing) of the pre-mRNA transcript into mature mRNA occurs in the nucleus.

Time spent: 0 sec
Subject:
Biology

QID: 400853
System:
Molecular Biology

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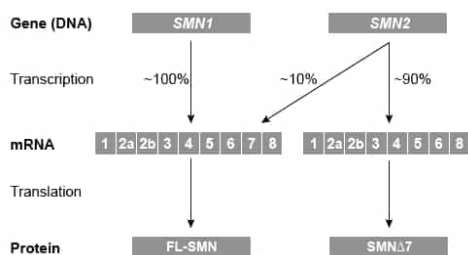


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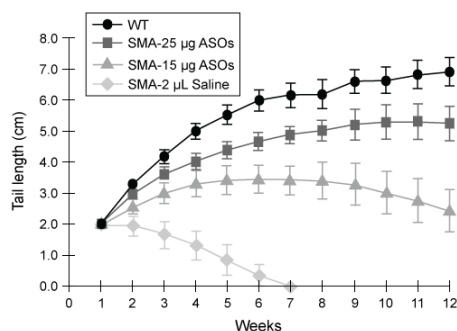


Figure 2 Tail length progression of WT mice compared to SMA mice treated with either saline or ASOs

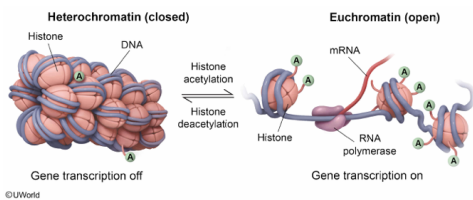
Valproic acid (VA) is a histone deacetylase inhibitor that has been investigated for SMA therapy. VA would most likely improve motor neuron survival by:

- ☒ A. increasing *SMN2* expression by making DNA more accessible. [61%]
- ☐ B. increasing *SMN2* expression by removing acetyl groups from histones. [11%]
- ☐ C. decreasing *SMN2* expression by modifying CpG sites in DNA. [4%]
- ☐ D. decreasing *SMN2* expression by inducing heterochromatin formation. [22%]

Correct

60% answered correctly

Explanation:



During the initial steps of **DNA packaging**, the DNA double helix wraps twice around a bead-like octamer complex known as the **histone core**. This DNA-histone complex, also called chromatin, can undergo specific chemical modifications that allow it to exist in two different conformations: the closed conformation (**heterochromatin**) and the open conformation (**euchromatin**).

Histones have tails (ie, N-termini protruding outward) accessible to modifying enzymes, which add or remove chemical groups to change chromatin structure and induce or repress gene expression. For example, histone acetylation, in which acetyl groups are added to histone tails, is catalyzed by the enzyme **histone acetylase**. Histone acetylation **promotes gene transcription** by loosening heterochromatin into euchromatin, making the DNA region readily accessible to the transcription machinery.

Conversely, **histone deacetylase** enzymes **downregulate gene expression** by removing acetyl groups from histones, which reverts euchromatin back to heterochromatin, restricting transcriptional access to specific genes. Histone **deacetylase inhibitors**, such as valproic acid (VA), **prevent removal of acetyl groups** by histone deacetylases.

As an SMA drug, VA would therefore *increase* acetylation of the *SMN2* gene. This would cause the genomic region containing *SMN2* to be **more accessible** to RNA polymerase and transcription factors, thereby *increasing SMN2* expression. Because *SMN2* produces *both* FL-SMN and SMNΔ7, *increased SMN2* expression would also *increase* FL-SMN expression, which would promote motor neuron survival.

(Choices B and D) Removal of acetyl groups from histones is carried out by histone deacetylase. An inhibitor of this enzyme (ie, VA) would *prevent (not promote)* removal of acetyl groups from histones. Therefore, histone deacetylase inhibitors would *prevent* heterochromatin formation (ie, *promote* euchromatic DNA structure), and lead to increased (*not decreased*) *SMN2* gene expression.

(Choice C) Addition of methyl groups to DNA, which often co-occurs with histone deacetylation, leads to the *silencing* of gene expression. **CpG sites**, or cytosine-guanine (CG) dinucleotide sequences, are methylated by the enzyme DNA methyltransferase, *not by histone deacetylase inhibitors*.

Educational objective:

Heterochromatin, or tightly packed chromatin, is composed of deacetylated histones (due to histone deacetylase activity) and is transcriptionally repressed. In contrast, euchromatin or relaxed chromatin is highly acetylated (due to histone acetylase activity) and transcriptionally active.

Time spent: 0 sec
Subject:
Biology

QID: 400854
System:
Molecular Biology

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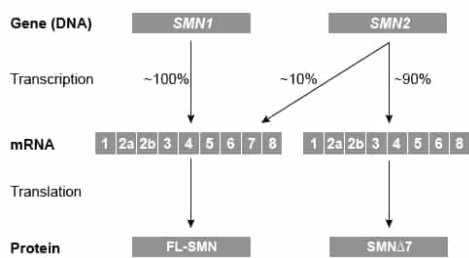


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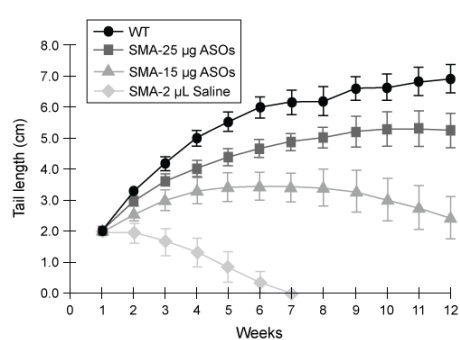


Figure 2 Tail length progression of WT mice compared to SMA mice treated with either saline or ASOs

Researchers have discovered that SMN plays a role in its own mRNA processing, and that FL-SMN enhances the expression of small nuclear RNAs (snRNA) required to produce mature FL-SMN mRNA. In motor neurons of patients with SMA, the feedback mechanism between SMN protein and SMN mature mRNA is shown in which northern blot sample?

☐ A.

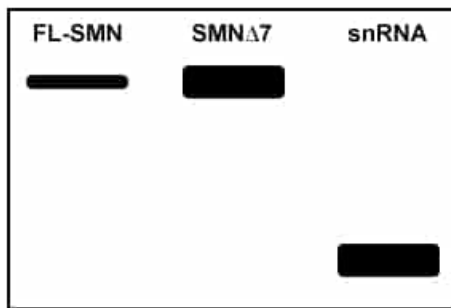
FL-SMN

SMN Δ 7

snRNA

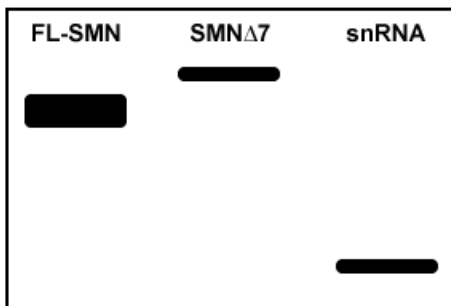
[33%]

☐ B.



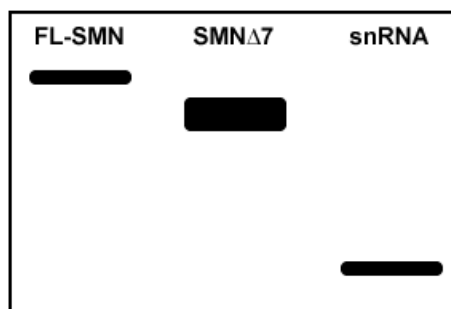
[11%]

☐ C.



[8%]

☒ D.

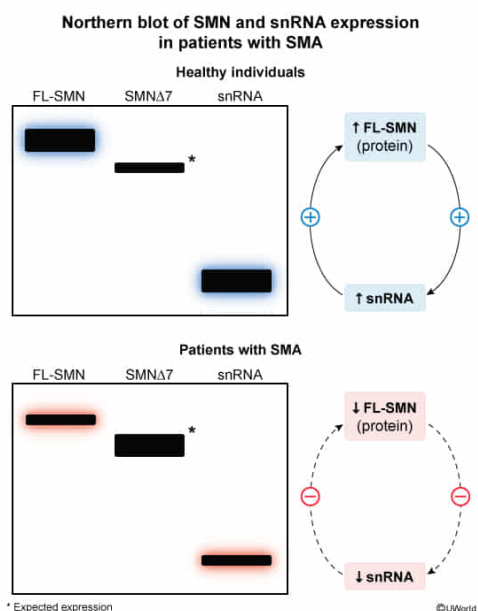


[46%]

Correct

47% answered correctly

Explanation:



Northern blotting is a laboratory technique used to detect and measure the concentration of specific **RNA molecules** within a sample. Isolated RNA strands are first separated by size via gel electrophoresis, after

微信Mai_ange

which RNA molecules are transferred from the gel to a blotting membrane. The membrane is incubated with a DNA or RNA probe labeled with a radioactive atom or fluorescent dye. This probe contains a sequence that is complementary to the target RNA, allowing the probe to hybridize, or bind, the specific RNA on the membrane. The labeled mRNA can then be visualized as bands on x-ray film, where **band intensity denotes quantity** of mRNA expression and **band location denotes size** (ie, smaller molecules appear lower than larger ones).

Patients with SMA, who only express *SMN2* gene products (of which FL-SMN is 10% of the total), will have **low concentrations** of **FL-SMN mRNA relative to SMNΔ7 mRNA**. Accordingly, a northern blot of RNA from motor neurons in patients with SMA would display a **higher** abundance of **SMNΔ7 mRNA** and a lower abundance of **FL-SMN mRNA**. The smallest mRNA molecule (snRNA) would travel farthest down the gel followed by SMNΔ7, which is shorter than FL-SMN mRNA because of the excluded exon 7. Only the blot shown in Choice D correctly displays these results.

(Choices A and C) In patients with SMA, truncated SMNΔ7 mRNA should have a *higher* concentration (ie, a more intense band) than both FL-SMN mRNA and snRNA because *SMN2* produces a majority (90%) of SMNΔ7 mRNA.

(Choice B) FL-SMN and SMNΔ7 mRNA are not equal in size (ie, SMNΔ7 is missing exon 7) and so should not be seen at the same position. In addition, for patients with SMA, the concentration of SMNΔ7 should be higher (*not* lower).

Educational objective:

Northern blots detect target RNA in a sample. The labeled RNA can be visualized as bands, where band intensity denotes the quantity of RNA expression and band position denotes the size (ie, smaller molecules appear lower than larger ones).

Time spent:0 sec

QID:400855

Subject:
Biology

System:
Molecular Biology